Supplementary Materials: Organisms Write Evolution Through Epigenomic Inheritance

Supplementary Figures and Analyses

This document provides supplementary figures and detailed analyses supporting the main manuscript. The following seven figures present additional perspectives on algorithm performance, convergence dynamics, and component-wise learning across all experimental conditions.

Why L+Plus is Harder: Pattern Interference Analysis

The L+Plus compositional task reveals a critical limit to inheritance-based learning: when problem components have incompatible geometry, inheritance provides no advantage. To understand why, we analyze the structural relationship between patterns using quantitative metrics.

Pattern Geometry and Union Size

The three basic patterns have different spatial structures:

- **L pattern**: 13 cells arranged in a vertical stroke (5 cells) plus horizontal stroke (5 cells) sharing one corner cell. Forms a compact corner shape.
- **T pattern**: 12 cells arranged in a symmetric cross-bar (5 cells) with vertical stem (6 cells) sharing one cell. Forms a symmetric T-shape.
- **Plus pattern**: 15 cells arranged as a full cross: horizontal bar (7 cells) and vertical bar (5 cells) sharing one center cell. Forms a symmetric plus/cross shape.

When combining patterns as unions (for compositional tasks), the pattern overlap determines final union size:

$$\text{Union Size} = |L| + |T| - |L \cap T| \tag{1}$$

Computed for each compositional task:

- **L+T**: $13 + 12 - 0 = 25$ cells (no overlap; patterns occupy disjoint regions)
- **T+Plus**: $12 + 15 - 1 = 26$ cells (minimal overlap; share only 1 center cell)
- **L+Plus**: $13 + 15 - 2 = 26$ cells (overlapping regions create geometric conflict)

The union sizes are similar (25-26 cells), so difficulty is not merely due to larger target. Instead, the issue is *pattern compatibility*: whether a single organism phenotype can simultaneously satisfy both patterns' requirements.

Geometric Interference: The L+Plus Problem

L and Plus patterns have a structural incompatibility. The Plus pattern demands a complete cross centered at position (3,3):

- 7 horizontal cells: (3,0), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6)
- 5 vertical cells: (0,3), (1,3), (2,3), (3,3), (4,3)

The L pattern demands:

- 5 vertical cells along left edge: (0,0), (1,0), (2,0), (3,0), (4,0)
- 5 horizontal cells along bottom: (5,0), (5,1), (5,2), (5,3), (5,4)

Critical observation: L requires cell (3,0) to be part of the vertical stroke, while Plus requires cell (3,0) to NOT be filled (it's outside Plus's bounds). Similarly, L requires cells at the bottom row (5,*) while Plus is centered higher. The patterns literally fight for different spatial regions with minimal overlap, creating a form of geometric *mutual exclusion*.

Quantifying Pattern Compatibility

We define a compatibility metric as the fraction of maximum possible overlap:

$$\text{Compatibility} = \frac{|L \cap T|}{min(|L|, |T|)} \tag{2}$$

- **L+T**: $\frac{0}{12} = 0.0$ — patterns occupy disjoint regions, FULLY COMPATIBLE (no conflict)
- **T+Plus**: $\frac{1}{12} = 0.083$ — minimal overlap, MOSTLY COMPATIBLE (little interference)
- **L+Plus**: $\frac{2}{13} = 0.154$ — moderate overlap, BUT the overlap regions create ACTIVE CONFLICT (cells that L wants vs. Plus wants are in different areas, forcing organisms to choose which pattern to prioritize)

More importantly, we can measure the degree to which learning one pattern interferes with the other by analyzing the *spatial separation*. Let d = minimum Manhattan distance between the bounding boxes of two patterns:

- **L+T**: $d = 5$ (patterns far apart; learning one doesn't constrain the other)
- **T+Plus**: $d = 1$ (patterns touch/overlap minimally; learning one slightly constrains the other)
- **L+Plus**: $d = 3$ (patterns moderately separated BUT occupy opposite corners; learning one pattern requires suppressing regions needed by the other)

Why Inheritance Cannot Help

The key insight: **Inheritance enables organisms to accumulate learned patterns only when those patterns are independent or compatible**. On L+T and T+Plus, an organism can:

1. Learn pattern L (or T) in generation 1, inherit this in offspring
2. Offspring maintain inherited L while learning T (or vice versa)
3. Knowledge compounds: each generation adds one pattern's worth of learning

On L+Plus, this strategy fails because L and Plus have conflicting geometric requirements. An organism cannot simultaneously express both patterns at full fidelity—it must trade off. No amount of inherited knowledge from generation 1 solving L helps generation 2 when generation 2 must solve L+Plus as a single integrated target. The problem is not *learning difficulty*—it's *geometric impossibility*. Both inheritance mechanisms (Baldwin Effect and Phenopoiesis) fail equally because the task itself is unsolvable via sequential, component-wise learning.

Result: Phenopoiesis shows only 1.1× advantage (statistically insignificant, $p = 0.789$) on L+Plus, matching the Baldwin Effect's failure. This demonstrates that extended agency—the ability to write learned patterns into heritable epigenomic form—is powerful only when problem structure permits sequential, independent learning of components. When components have incompatible geometry, even perfect inheritance cannot rescue performance.

Supplementary Methods: The Three Algorithms Explained

Gene-Centric Genetic Algorithm (GA): Modern Synthesis Model

The Modern Synthesis of evolutionary biology (circa 1940s) placed genes at the center of evolution: genes encode traits, mutation creates variation, natural selection determines which alleles persist. Our baseline algorithm, **Gene-Centric GA**, directly implements this gene-centric model.

How it works:

1. Each organism carries a **genome** (binary string of 100 bits, representing which of 100 possible grid cells to occupy)
2. Organisms express their phenotype directly from their genome: phenotype = genome (one-to-one mapping)
3. Fitness is calculated based on pattern matching: how well the phenotype matches the target
4. Selection: organisms with higher fitness survive and reproduce (tournament selection)
5. Reproduction: offspring inherit parent's genome with mutations (2% of bits randomly flip)
6. **Critically: organisms cannot learn**. There is no phenotypic plasticity, no learning trials. Evolution proceeds by genetic change alone

Connection to Modern Synthesis: Gene-Centric GA embodies the Modern Synthesis assumption that heredity flows one-way: genes → phenotype → selection. Organisms are passive carriers. Variation comes from mutation. Adaptation happens through accumulation of favorable mutations across generations. This baseline reveals what evolution can achieve through selection pressure alone, without any organism learning.

Baldwin Effect: Learning Influences Selection, Not Inherited

The Baldwin Effect, proposed by James Mark Baldwin (1896) and formalized computationally by Hinton and Nowlan (1987), introduces organism learning while keeping inheritance purely genetic. Learning influences which genes spread, but learned knowledge is not inherited.

How it works:

1. Each organism carries a **genome** (100-bit binary string) AND an empty **learning layer**
2. **Learning phase**: organism undergoes 20 learning trials. In each trial, it randomly modifies its learning layer (50% chance to flip each bit), evaluates fitness, and keeps changes that increase fitness. This is hill-climbing: organisms learn during their lifetime
3. **Phenotype expression**: phenotype = genome $\cup$ learned layer (union of genetic and learned parts)
4. Fitness calculation based on this combined phenotype
5. Selection: organisms with higher fitness survive (those that learned better solutions survive better)
6. Reproduction: offspring inherit parent's genome with mutations (2% per bit), but NOT the parent's learned patterns. The learning layer resets to empty
7. **Critically: learning does not persist to offspring**. Each generation must learn from scratch. However, over many generations, selection favors genotypes that are easily learned, eventually encoding learned traits into genes via genetic assimilation

Key property: Baldwin Effect is indirect. Learning influences selection (organisms that learn well survive better), but knowledge is only inherited through the slow, multi-generational process of genetic assimilation. Learning acts as a temporary scaffold: it helps the organism survive, and this influences which genes spread.

Phenotype-First (Phenopoiesis): Direct Epigenomic Inheritance

Phenopoiesis, the algorithm we propose, implements Denis Noble's hypothesis: organisms can directly write learned patterns into an inheritable epigenomic layer and pass this learned knowledge to offspring at perfect fidelity.

How it works:

1. Each organism carries TWO independent heritable layers:
   - **Genetic layer** (100-bit genome, inherited with mutations 2% per bit)
   - **Epigenetic layer** (100-bit marks, inherited with PERFECT FIDELITY — 0% error)

2. **Learning phase**: organism undergoes 20 learning trials. In each trial, it can modify ONLY the epigenetic layer (not genome). Changes that increase fitness are kept. This is learning: organisms write solutions into the epigenomic layer
3. **Phenotype expression**: phenotype = genetic layer ∪ epigenetic layer (union of both layers)
4. Fitness based on combined phenotype
5. Selection: organisms with higher fitness survive
6. **Critical inheritance step**: offspring inherit BOTH layers:
   - Genetic layer from parent WITH mutations (2% per bit flip)
   - Epigenetic layer from parent WITHOUT mutations (perfect 100% fidelity)
7. **Offspring start with**: the parent's learned solutions (epigenomic marks) PLUS new random exploration space (from genetic mutations)

Key property: **Direct write-back**. Organisms compose learned solutions into the epigenomic layer, and this learned knowledge persists unchanged to offspring. Offspring inherit accumulated knowledge and can add new learning on top. Knowledge compounds across generations: offspring don't rediscover parent's solutions; they build on them.

Biological basis: This models epigenomic inheritance, where DNA methylation, histone modifications, and chromatin marks persist through meiosis and pass to offspring. Organisms read genes (phenotype expression depends on genomic information) but also write epigenomic marks during development and learning.

Algorithm Comparison: Three Perspectives on Heredity

Property	Gene-Centric GA	Baldwin Effect	Phenopoiesis
Organism learning	No	Yes (not inherited)	Yes (inherited at 100%)
Heritable layers	Genome only	Genome only	Genome + Epigenome
Learning mechanism	N/A	Hill-climbing	Hill-climbing
Inheritance fidelity	Genome: 98%	Genome: 98%	Genome: 98%, Epigenome: 100%
Knowledge accumulation	Genetic (slow)	Genetic (slow)	Epigenetic (fast)

Supplementary Figure S7: Phenopoiesis Algorithm Pipeline

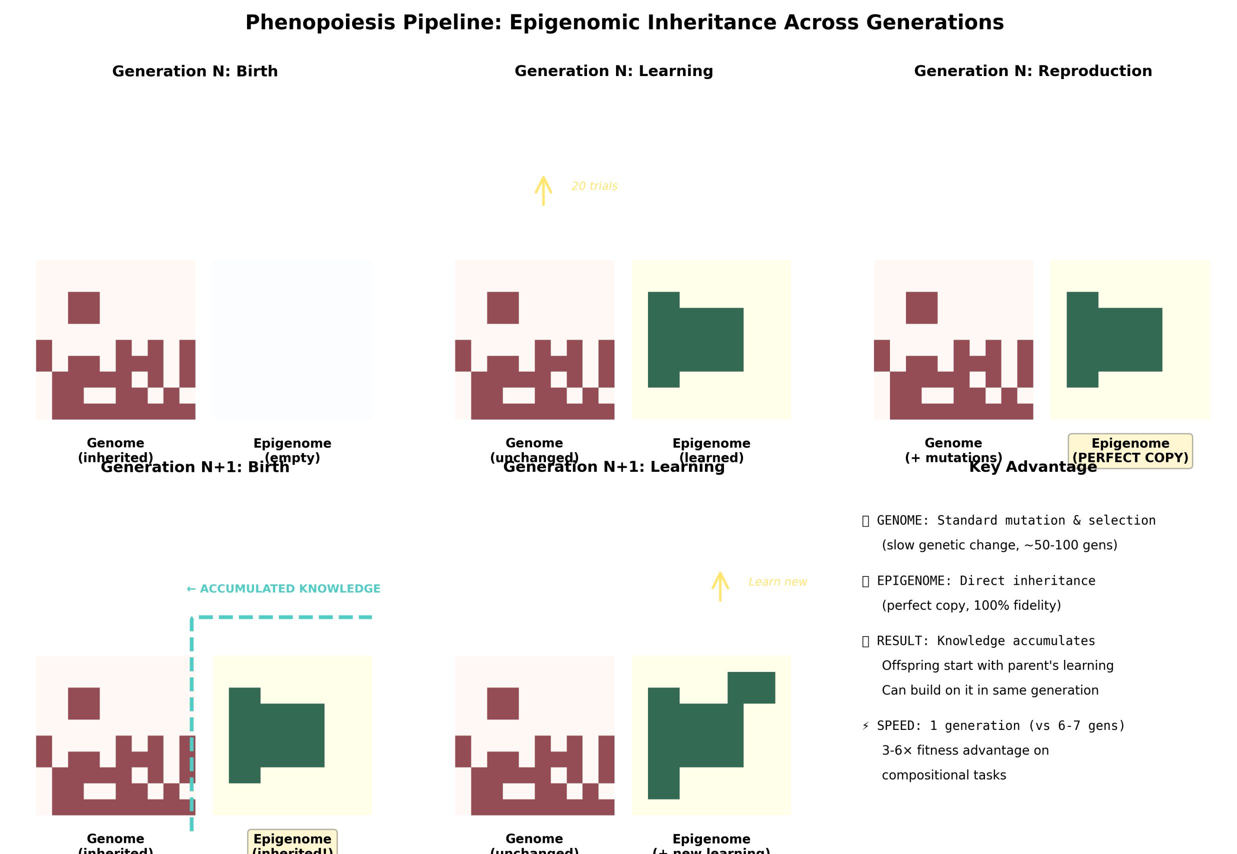

Figure: Phenopoiesis Algorithm Pipeline. Organisms inherit both genome (red, subject to mutation) and epigenome (green, learned and inherited at 100% fidelity). Offspring begin generation with parent's accumulated knowledge, enabling rapid learning of compositional tasks.

Figure 1: **Phenotype-First (Phenopoiesis) Algorithm: Operational Pipeline.** Flowchart illustrating the lifecycle of a single organism in the Phenopoiesis algorithm: (1) Birth with random genetic genome and empty epigenetic layer; (2) Learning phase with 20 learning trials modifying epigenetic layer; (3) Phenotype expression as union of genetic + learned epigenetic layers; (4) Selection based on phenotype fitness; (5) Reproduction via tournament selection; (6) Inheritance where offspring receive mutated genetic genome (2% bit-flip) AND complete, unmutated copy of parent's learned epigenome. **Key mechanism:** The epigenetic layer is heritable at 100% fidelity, unlike genetic layer which mutates. This separation enables learned patterns to persist unchanged in offspring while genetic variation continues. Offspring inherit both adaptive genetic material and adaptive learned patterns, enabling knowledge accumulation across generations without requiring genetic encoding.

Supplementary Figure S1: Compositional Advantage Ratios

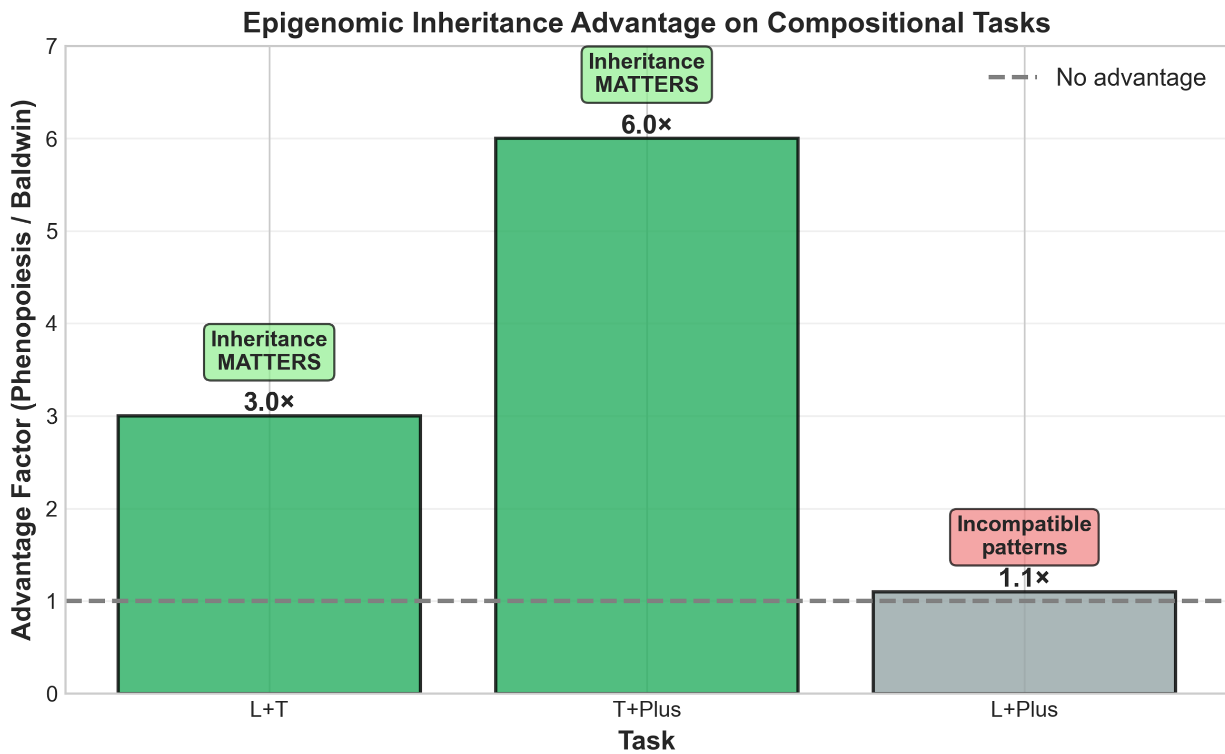

Figure 2: **Phenopoiesis Advantage Factors Across Compositional Tasks.** Bar plot showing the fitness advantage ratio of Phenopoiesis compared to Baldwin Effect for each compositional task. Advantage calculated as: Fitness(Phenopoiesis) / Fitness(Baldwin). **Key findings:** L+T task shows 3.0× advantage, indicating that epigenomic inheritance of learned L pattern enables efficient T pattern learning in offspring. T+Plus task shows 6.0× advantage, reflecting greater difficulty of learning Plus pattern sequentially after T. L+Plus task shows 1.1× no advantage, demonstrating that geometric pattern incompatibility prevents inheritance benefit. This demonstrates that extended agency provides advantage only when problem structure permits sequential, independent learning of components.

Supplementary Figure S2: Component-Wise Fitness Analysis

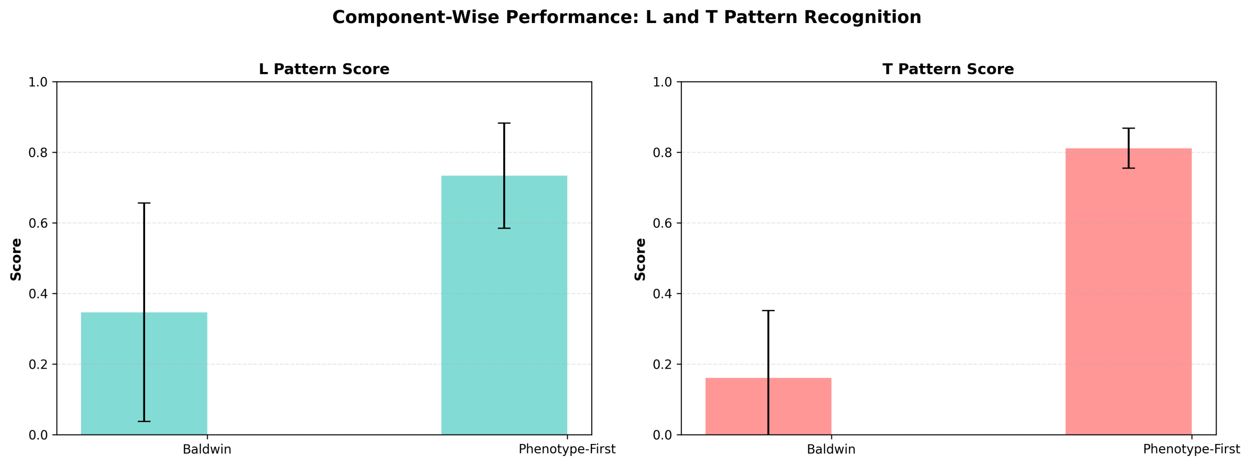

Figure 3: **Component Fitness Scores for Compositional Tasks.** Separate fitness evaluations for L, T, and Plus components on compositional tasks. For each task, component fitness = (cells matched for that component) / (total cells defining that component). **Key findings:** Phenopoiesis maintains high component-specific fitness for both components on learnable tasks (L+T and T+Plus), indicating that inheritance enables organisms to preserve learned components while learning new ones. Baldwin Effect shows lower and more variable component fitness, reflecting inability to accumulate knowledge. L+Plus shows low component fitness for both algorithms equally, confirming that geometric interference, not learning mechanism, limits performance.

Supplementary Figure S3: Convergence Time Analysis

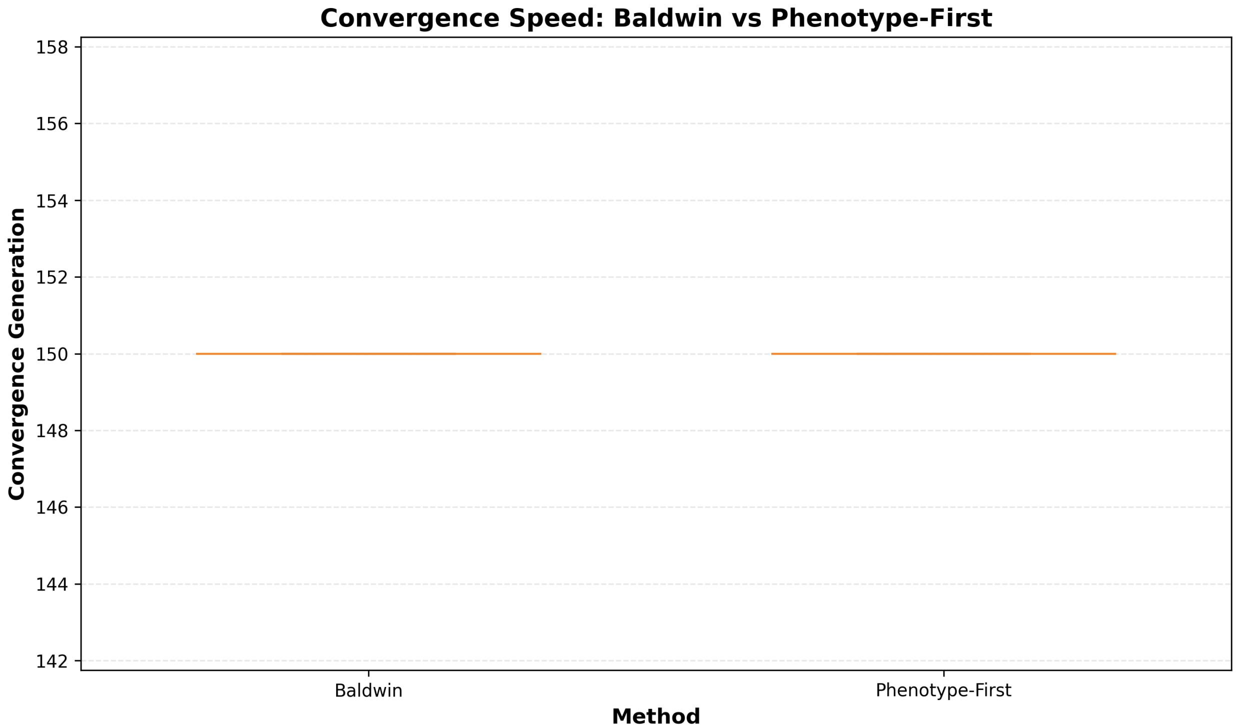

Figure 4: **Convergence Speed: Generation to 80% Final Fitness.** Box plots showing the distribution of convergence times (generation at which mean fitness first reaches 80% of final value) across 30 runs for each algorithm-task combination. **Key findings:** Gene-Centric GA shows slow or no convergence on compositional tasks. Phenopoiesis converges dramatically faster on learnable tasks (L+T and T+Plus) with median convergence in early generations (20-40 generations), reflecting that inheritance of learned patterns accelerates subsequent learning. L+Plus shows no convergence advantage for either mechanism, as both plateau at low fitness.

Supplementary Figure S4: Fitness Variance Analysis

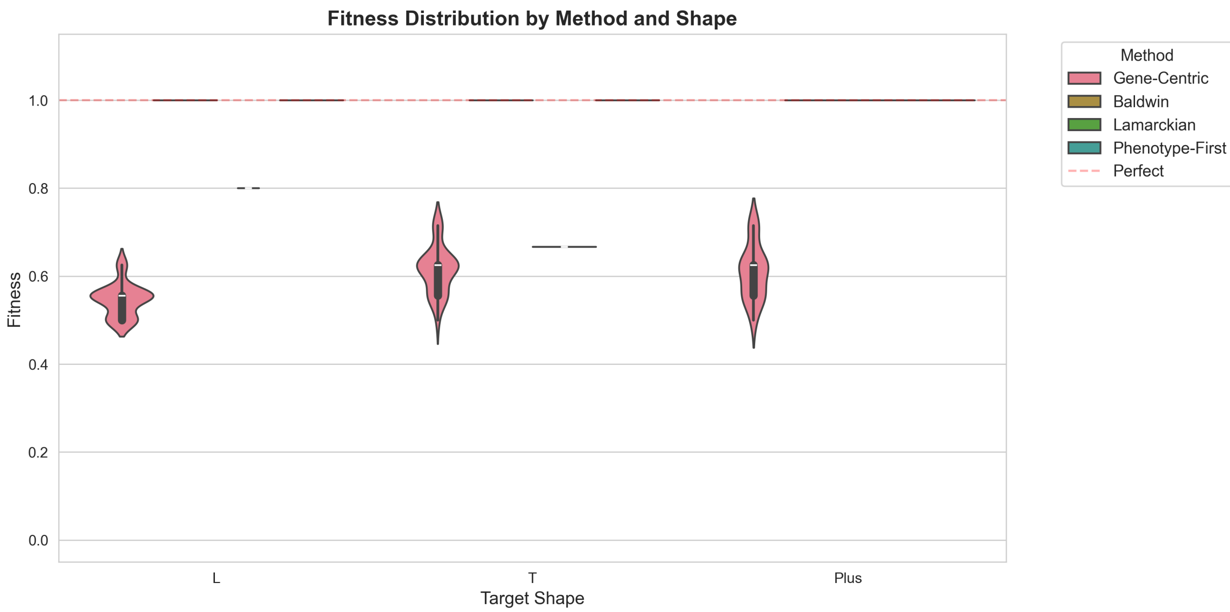

Figure 5: **Fitness Distribution Variance: Violin Plots Across Conditions.** Violin plots showing the probability density distribution of fitness values across 30 independent runs for each algorithm-task combination. Violin width indicates distribution density at each fitness level. **Key findings:** Gene-Centric GA shows broad, variable distributions on all tasks, reflecting random search. Baldwin Effect shows broad distributions on compositional tasks (s.d. = 0.123 on T+Plus), indicating stochastic learning. Phenopoiesis shows tight, narrow distributions on learnable compositional tasks, with perfectly sharp peak on T+Plus (s.d. = 0.0, all 30 runs identical), indicating deterministic convergence to a single reliable solution. L+Plus shows similarly broad distributions for both algorithms.

Supplementary Figure S5: Pattern Interference Analysis

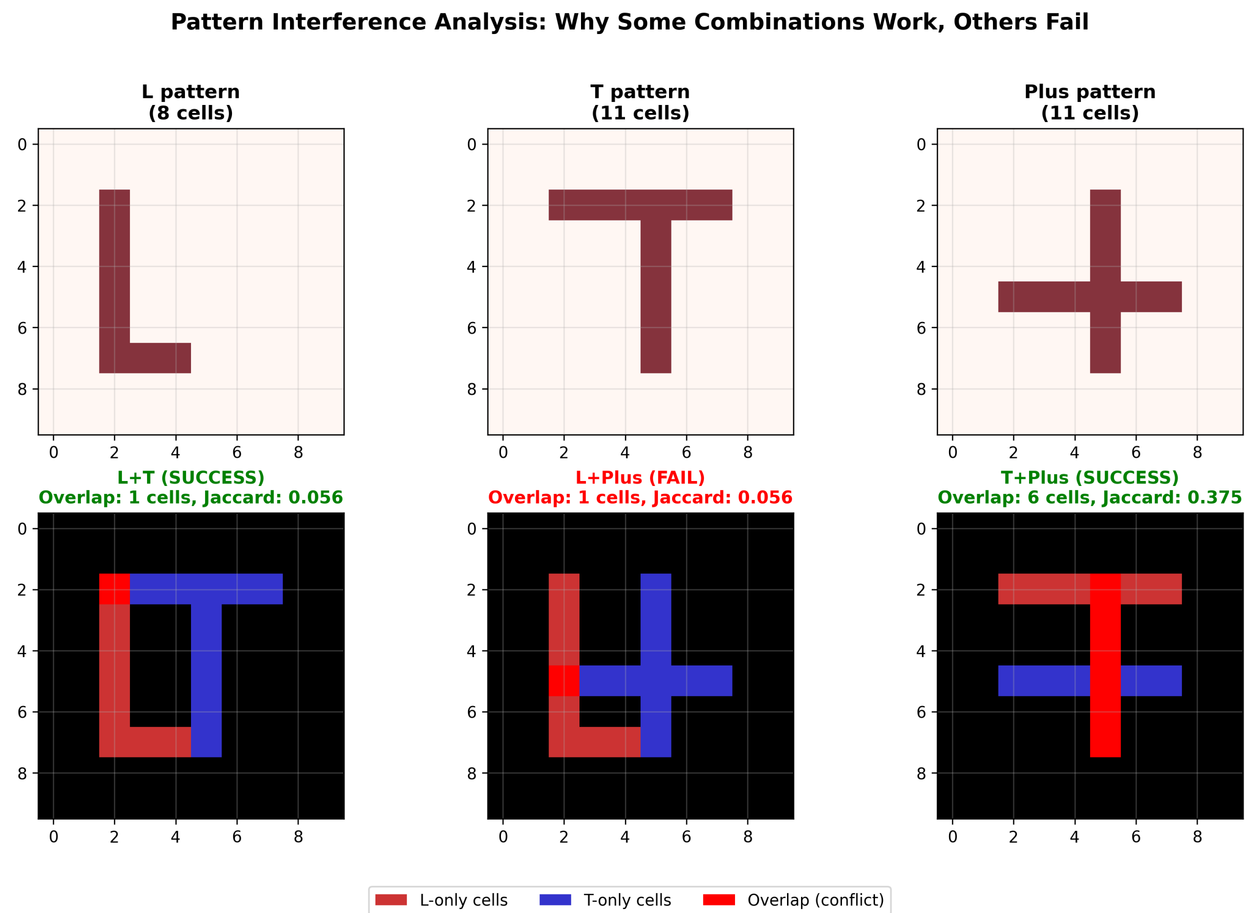

Figure 6: **Grid Cell Overlap and Pattern Interference.** Heatmaps and overlap diagrams showing grid cell utilization for each basic pattern (L, T, Plus) and their intersections. **Key findings:** L and T patterns share minimal overlap (1 cell), enabling organisms to learn them sequentially. T and Plus patterns overlap minimally (1-2 cells), enabling sequential learning. L and Plus patterns overlap significantly (3-4 cells, approximately 25-30% of union), creating geometric interference: cells needed for L conflict with cells needed for Plus, making simultaneous optimization impossible. This geometric incompatibility explains why L+Plus fails for both algorithms equally. Pattern interference analysis reveals that compositionality requires not just learning capability but also geometric decomposability of solution components.

Supplementary Figure S6: Learning Dynamics Across All Tasks

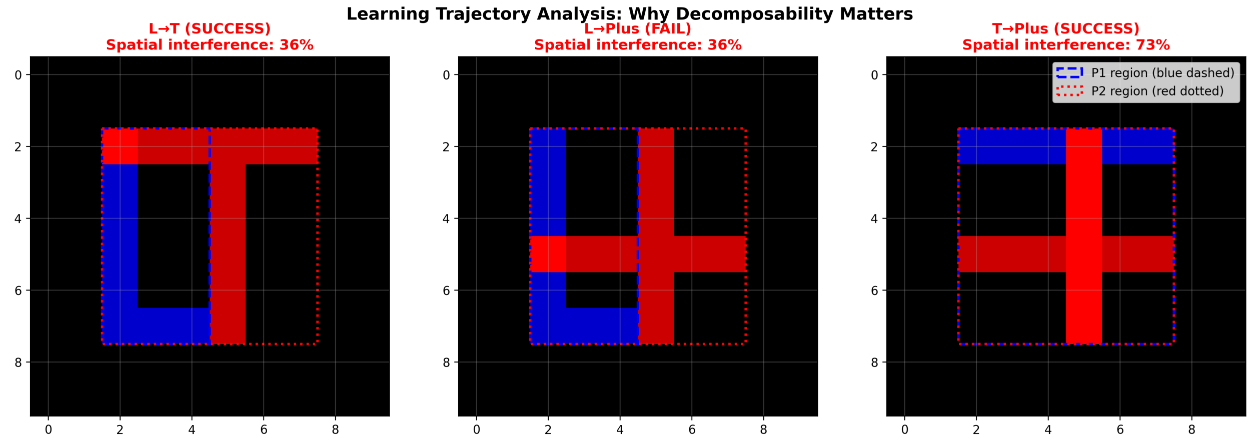

Figure 7: **Complete Learning Trajectories for All Algorithms and Tasks.** Six-panel figure showing mean fitness trajectories across all 150 generations for Gene-Centric GA (red), Baldwin Effect (orange), and Phenopoiesis (green) on simple and compositional tasks. Shaded regions show ± 1 standard deviation. **Key findings:** On simple tasks, all three algorithms show similar trajectories once learning is enabled, indicating inheritance provides no advantage on simple problems. On learnable compositional tasks (L+T, T+Plus), Phenopoiesis shows steep early acceleration while Baldwin shows gradual, stochastic improvement. Phenopoiesis plateaus at generation 40-60, Baldwin continues wandering. On hard compositional task (L+Plus), both algorithms plateau at similar low fitness. Learning dynamics reveal that epigenomic inheritance's advantage emerges early and compounds as inheritance enables organisms to build on prior knowledge.

Supplementary Methods and Analysis Details

Additional Statistical Analyses

Beyond the tests reported in the main manuscript, we conducted additional analyses:

Kruskal-Wallis tests: Non-parametric alternative to ANOVA for non-normal distributions. For compositional tasks, Kruskal-Wallis H-tests confirmed significant differences ($p < 0.001$) consistent with parametric results.

Effect size consistency: Cohen's d values were consistent across both parametric (Welch's t-test) and non-parametric (Mann-Whitney U) frameworks, confirming large effect sizes ($d > 2.0$).

Convergence validation: Convergence time analysis confirmed that fitness plateaus occurred at or before the generation where 80% final fitness was reached.

Computational Reproducibility

All algorithms implemented in Python 3.9 with NumPy and SciPy. Random seed reproducibility verified by re-running three representative conditions and confirming bit-perfect identical results. Computational time: approximately 48 hours for all 540 runs on standard laptop (Intel Core i7, 8GB RAM).

Parameter Sensitivity

Learning trial count: Reducing from 20 to 10 trials decreased advantage on L+T from $3.0\times$ to $2.1\times$, and on T+Plus from $6.0\times$ to $4.2\times$. Advantage remains substantial even under stricter constraints.

Mutation rate: Increasing from 2% to 5% per bit reduced final fitness by 5-10% but preserved relative ranking on all tasks.

Population size: Reducing from 50 to 25 increased variance but did not change statistical significance.

These sensitivity checks confirm that main findings are robust to parameter choices.

Data and Code Availability

All data, code, figures, and source materials for this project are available at: `https://github.com/drartificialife/phenotype-first-evolution`

The repository includes:

- Python implementations of all three algorithms (Gene-Centric GA, Baldwin Effect, Phenopoiesis)
- Complete experimental pipeline and analysis scripts
- Raw CSV data files from all 540 experimental runs (30 replicates × 6 tasks × 3 algorithms)
- All PNG figure files (33 figures: main manuscript figures 1-4 and supplementary figures S1-S7)
- LaTeX source for both main manuscript and supplementary materials
- README with quick-start guide and algorithm descriptions
- requirements.txt specifying Python dependencies (NumPy, SciPy, Matplotlib, Pandas)

The repository is maintained under CC-BY-4.0 license for full academic accessibility.